Cover Page

By signing below, each group member confirms that:

- I have read and understood the assignment guidelines and grading policy.
- I contributed to this assignment as indicated, and I am responsible for the work submitted under my name.
- I have had **no exchanges of assignment-related material** with anyone outside my group. *(Examples: sharing code, discussing solutions, debugging together)*
- **If AI tools were used**, I have listed them below together with their specific purpose. If none were used, I write: "None".

Group Members and Contributions (Total = 100%)

1. Name: _____
   Student number: _____
   AI Tools Used: _____
   Signature: _____

2. Name: _____
   Student number: _____
   AI Tools Used: _____
   Signature: _____

3. Name: _____
   Student number: _____
   AI Tools Used: _____
   Signature: _____

Designing a Droid–Assisted Pickup and Delivery System on a Time–Space Network

Autonomous delivery droids are increasingly deployed to support urban logistics operations. Droids are capable of performing local pickup and delivery tasks but possess only limited operating range; once their available energy is depleted, they must be collected by a truck and transported before they can be used again. Trucks move freely across the service region, carry parcels as well as multiple droids, and redeploy droids at suitable locations. The objective of this assignment is to design a service network that coordinates truck activities and droid excursions so that all known pickup and delivery requirements are satisfied.

Spatial and temporal structure. The service region is partitioned into a set of hexagonal zones of equal area, denoted by Z. Each zone has up to six neighbors; let $N(z) \subseteq Z$ denote the set of zones adjacent to $z \in Z$. The depot is located at the centroid of a designated depot zone $z_0 \in Z$. Travel time from the centroid of a zone to the centroid of any adjacent zone is one time interval. Time is discretized into equal-length periods indexed by $t \in \{1, \ldots, T\}$, where T denotes the final period of the planning horizon. We interpret period t as the time interval $[t, t{+}1)$, so a point (z, t) represents being in zone z during period t. Vehicles start at the beginning of period 1 and must be back at the beginning of period T. In each period, a vehicle either remains in its current zone or travels to an adjacent zone, and its location is updated for the next period accordingly. Either vehicle type may remain idle for any number of periods, consuming time without expending distance or energy.

Vehicle movement and service rules. A truck requires one period to travel to an adjacent zone and is not subject to any energy constraints. To perform a pickup or delivery in a zone, a truck must remain stationary in that zone for one full period, during which the loading or unloading operation is completed. A truck possesses two independent capacities: one limiting the number of parcels it may carry and one limiting the number of droids it may transport.

A droid likewise requires one period to move to an adjacent zone. Each droid has a parcel capacity and an energy state. While operating independently, a droid expends one unit of energy for every movement it performs and for every pickup or delivery action it performs; waiting consumes no energy. A droid may perform pickup and/or delivery actions while located in its current zone during period t, subject to a per-period service cap: during any period t, a droid may serve at most $S^{\max}$ requests (pickups and/or deliveries in total) in its current zone.

Transfers and service actions may occur during the same period when vehicles are co-located in the same zone. For example, if two vehicles are together in zone z during period t, a parcel may be transferred between them during period t, and the receiving vehicle may complete a service in zone z during that same period.

Each droid begins the planning horizon fully charged. While operating independently, a droid cannot replenish energy on its own. A droid may be collected by a truck only when both vehicles are in the same zone during the same period; once collected, it may be transported and redeployed later in the horizon. When a droid is carried by a truck, it does not consume energy and its location coincides with the carrying truck's location for the periods in which it remains onboard.

Excursions. For the purposes of this assignment, the interval between the moment a droid departs from a truck (i.e., leaves a truck to operate independently) and the moment it next rejoins any truck (i.e., boards a truck) is termed an *excursion*. Range feasibility will be expressed in terms of limits on the number of energy-consuming actions a droid may

execute during such excursions; in the scenarios below, this will be specialized first to a no-recharging setting and later extended to allow recharging.

Pickup and delivery requests. The system must fulfill a known set of service requests. A pickup request is defined by a zone and a release time; service may occur at any period at or after its release time, and the collected parcel must eventually be transported to the depot. A delivery request is defined by a zone and a due time; service may occur at any period up to and including its due time, and every delivery originates from the depot. Unless otherwise stated in the dataset, each delivery parcel is available at the depot at the start of the horizon. Either a truck or a droid may perform a request, provided it is in the required zone at an admissible service time and the relevant capacity constraints are respected.

Parcels carried by a droid remain with that droid until they are delivered, returned to the depot, or transferred to another vehicle. Parcels carried by a truck remain on board until delivered, returned to the depot, or transferred. The scenarios below specify which request types and which transfer options are active in each case.

Capacities, loading, and transfers. Each request corresponds to exactly one parcel and consumes one unit of parcel capacity while it is carried. Trucks have a parcel capacity and a droid-carrying capacity; droids have a parcel capacity. Truck parcel capacity applies to all parcels onboard the truck, regardless of origin or which vehicle will ultimately serve them. A droid may board or leave a truck only when both are in the same zone during the same period, and truck droid capacity must never be violated.

Parcel transfers are permitted only when the involved vehicles are co-located in the same zone during the same period. All transfers must respect the resulting parcel capacities of the involved vehicles. Delivery parcels are available at the depot in period 1 and may be loaded and transferred at the depot in period 1 without consuming a time period.

In all settings, the objective is to minimize total transportation cost, including a fixed cost for deploying a truck. The variable cost for a droid is proportional to its energy consumption (one unit per movement and one unit per service action). The variable cost for a truck is based on its traversed travel and service activity; whether staying/waiting in a period incurs cost when no service is performed is specified by the arc-cost data provided with the dataset.

The subsequent tasks in this assignment ask you to construct optimization models that enforce these requirements under progressively richer operational assumptions.

Assignment Tasks

For each scenario, formulate a complete optimization model on the given time–space network. Your formulation must define all decision variables, constraints, and the objective function in a way that is directly implementable on the provided dataset. Implement each model and report the requested outputs.

1. **Scenario 1: Baseline Model – Truck–Droid Deliveries with Fixed Droid Range.**

   In this scenario, only *delivery* requests are present. Trucks and droids move on the shared time–space network and may wait. Parcels originate at the depot at the start of the horizon and may be carried by trucks and/or droids; truck–droid parcel transfers are allowed only when the two vehicles are co-located in the same zone during the same period. Droids do not recharge: each droid has a fixed total energy

budget R over the full horizon, where one unit of energy is consumed for each ground movement and for each delivery action performed by the droid; waiting and riding on a truck consume no energy. During any period t, a droid may serve at most $S^{\max}$ requests in its current zone. All trucks and droids start at the depot and must return to the depot by period T, and no parcels may remain on any vehicle at the end of the horizon.

Report. Provide (i) the number of trucks deployed and the resulting truck and droid time–space trajectories, (ii) which vehicle ultimately performs each delivery request (and when), and (iii) a figure illustrating representative truck routes and droid excursions.

2. **Scenario 2: Deliveries with Droid Recharging and Multiple Excursions.**

   Scenario 2 uses the same delivery requests, fleets, transfer rules, capacities, and service rules as Scenario 1. The only change is the droid range semantics: whenever a droid rides on a truck for any period, it is considered fully recharged; thus, each time the droid next operates independently (i.e., after leaving a truck), its available energy is reset to the full range R. Equivalently, every excursion (from leaving a truck to the next time it boards a truck) has its own budget of R energy-consuming actions. All other requirements, including the per-period droid service cap $S^{\max}$ and the end-of-horizon emptiness condition, remain in effect.

   Report. Present the revised formulation with a clear description of how the Scenario 1 energy constraint is replaced by excursion-based constraints. Report (i) truck and droid trajectories, (ii) the number and timing of excursions per droid, and (iii) a figure illustrating at least one droid executing multiple excursions and indicating where range resets occur.

3. **Scenario 3: Pickup and Delivery with Truck–Droid and Droid–Droid Transfers.**

   Scenario 3 extends Scenario 2 by adding *pickup* requests and allowing *droid–droid* parcel transfers in addition to truck–droid transfers. Pickup requests have release times and generate a parcel when served; each pickup parcel must be transported to the depot by period T. Deliveries remain as before (each delivery parcel originates at the depot and must be served by its due time). Parcels may be carried by trucks and droids subject to capacities, and may be transferred between any two co-located vehicles when such transfers are enabled (truck–droid and droid–droid). The excursion-based recharging behavior from Scenario 2 remains unchanged, and the droid per-period service cap $S^{\max}$ applies to the total number of pickup and delivery actions performed by that droid in that period. As in the previous scenarios, all vehicles must return to the depot by T, and no parcels may remain on any vehicle at the end of the horizon.

   Report. Provide the full formulation including explicit parcel movement and transfer constraints. Report (i) the integrated plan of trucks and droids for both pickups and deliveries, and (ii) illustrations highlighting truck–droid and droid–droid transfer points in the time–space network.

Grading Rubric (For Students)

Your grade is based primarily on (i) the correctness of your mathematical models and (ii) whether your implementation runs correctly on instances in the provided data format. Clear explanations and figures are required, but they do not substitute for correctness.

Important. Alternative valid formulations are welcome. You will not be penalized for using a different modeling approach (e.g., arc-based vs. node-based flow) as long as it enforces the stated rules exactly and is clearly explained and implemented.

Grading Rubric (For Students)

Your work is evaluated on (i) correctness of the optimization *formulation*, (ii) correctness and robustness of the *implementation* on the provided data format, and (iii) clarity of *reporting*. Alternative valid formulations are accepted if they correctly capture the scenario rules and are clearly justified.

1. Scenario 1 (30%)

Formulation (18%): a correct and complete model for Scenario 1 consistent with the stated system rules.
Implementation (9%): runnable notebook that produces solutions consistent with the model and runs on additional instances with the same format.
Reporting (3%): clear presentation of key outputs and at least one informative figure.

2. Scenario 2 (25%)

Formulation (15%): correct extension of Scenario 1 capturing excursion-based range recharging.
Implementation (7%): correct behavior on the provided instance and additional instances.
Reporting (3%): clear explanation of what changes from Scenario 1 and a figure illustrating excursions/resets.

3. Scenario 3 (35%)

Formulation (22%): correct extension of Scenario 2 incorporating pickups and parcel transfers (including droid–droid transfers when enabled).
Implementation (10%): correct handling of pickups, transfers, and depot return requirements on the provided instance and additional instances.
Reporting (3%): clear figures and concise description of how transfers and pickup returns are realized.

4. Comparative summary (10%)

Analysis (10%): correct comparison of outcomes across scenarios (feasibility, cost, and operational structure), supported by reported results.

General expectations and grading policy

- **Correctness dominates:** incorrect models cannot receive high marks regardless of presentation.

- **Reproducibility:** the notebook must run end-to-end from a fresh kernel using the provided data format; hard-coded instance-specific fixes will be penalized.
- **Non-running code:** receives 0 for implementation components; the written formulation can still receive credit if correct and complete.
- **AI disclosure:** if AI tools are used, list them and their purpose on the cover page; grades depend on correctness and clarity.

Sets and Indices

- Z: set of physical zones with z_0 being the depot
- $T = \{1, 2, 3, \ldots, T\}$: set of time periods
- $N = Z \times T$: set of time-space nodes
- A: set of travel arcs in the time space network. Travel arcs $(z \to z')$ are defined as: $((z, t), (z', t + 1))$. Waiting at node z for one time-period can be written as $((z, t), (z, t + 1))$
- K: set of trucks, indexed by k
- D: set of droids, indexed by d
- P: set of parcel request, indexed by p
- $\delta^+(i)$: set of outgoing arcs from node $i \in N$
- $\delta^-(i)$: set of incoming arcs from node $i \in N$

Parameters

- c_{ij}^{truck}: cost of truck to traverse arc $(i, j) \in A$
- c_{ij}^{droid}: cost of droid to traverse arc $(i, j) \in A$. This is equal as the energy consumption of the droid
- f_k: fixed cost to deploy truck $k \in K$
- e_{ij}^{droid}: energy consumed by droid on arc (i, j), where:
  - $e_{ij}^{\text{droid}} = 1$ if arc (i, j) is a travel arc
  - $e_{ij}^{\text{droid}} = 0$ if arc (i, j) is a waiting arc
- Q_k^{truck}: parcel capacity of truck $k \in K$
- Q_k^{droid}: droid carrying capacity of truck $k \in K$
- $Q_d^{\text{droid-parcel}}$: parcel capacity of droid $d \in D$
- R_d: energy capacity of droid $d \in D$
- $S^{\max}$: max services per period
- z_p: delivery zone for request $p \in P$
- $\bar{t}_p$: due-time for request $p \in P$

With these parameters, the set $N_p \subseteq N$ can be made which is the set of space-time nodes where request p can be served: $N_p = \{(z_p, t) : t \leq \bar{t}_p\}$

Decision Variables

- $y_{ij}^k \in \{0, 1\}$: binary variable equals 1 if truck k uses arc $(i, j) \in A$, 0 otherwise
- $u_{ij}^d \in \{0, 1\}$: binary variable equals 1 if droid d uses arc $(i, j) \in A$ independently, 0 otherwise
- $w_k \in \{0, 1\}$: binary variable equals 1 if truck k is deployed
- $x_{ij}^{pk} \in \{0, 1\}$: flow of parcel request p on arc (i, j) carried by truck k

- $\xi_{ij}^{pd} \in \{0,1\}$: flow of parcel request p on arc (i,j) carried by droid d
- $s_i^{pk} \in \{0,1\}$: binary variable equals 1 if request p is served by truck k at node $i \in N_p$, 0 otherwise
- $\sigma_i^{pd} \in \{0,1\}$: binary variable equals 1 if request p is served by droid d at node $i \in N_p$, 0 otherwise
- $b_{ij}^{dk} \in \{0,1\}$: binary variable equals 1 if droid d is carried by truck k on arc (i,j), 0 otherwise
- $\alpha_i^{pkd} \in \{0,1\}$: binary variable equals 1 if parcel p is transferred from truck k to droid d at node i, 0 otherwise
- $\beta_i^{pkd} \in \{0,1\}$: binary variable equals 1 if parcel p is transferred from droid d to truck k at node i, 0 otherwise

Objective Function

$$\min \sum_{k\in K} f_k w_k + \sum_{k\in K}\sum_{(i,j)\in A} c_{ij}^{\text{truck}} y_{ij}^k + \sum_{d\in D}\sum_{(i,j)\in A} c_{ij}^{\text{droid}} u_{ij}^d$$

In this:

- $\sum_{k\in K} f_k w_k$: is the fixed cost of deploying a truck
- $\sum_{k\in K}\sum_{(i,j)\in A} c_{ij}^{\text{truck}} y_{ij}^k$: the variable cost of truck operation
- $\sum_{d\in D}\sum_{(i,j)\in A} c_{ij}^{\text{droid}} u_{ij}^d$: the variable cost of droid operation (proportional to its energy consumption)

Constraints

$$\sum_{j\in\delta^+(i)} y_{ij}^k - \sum_{j\in\delta^-(i)} y_{ji}^k = \begin{cases} w_k & \text{if } i=(z_0,1) \quad \text{(start depot)} \\ -w_k & \text{if } i=(z_0,T) \quad \text{(end depot)} \\ 0 & \text{otherwise} \end{cases} \quad \forall k\in K, \forall i=(z,t)\in N \tag{1}$$

$$\sum_{j\in\delta^+(i)} u_{ij}^d - \sum_{j\in\delta^-(i)} u_{ji}^d + \sum_{k\in K}\left(\sum_{j\in\delta^+(i)} b_{ij}^{dk} - \sum_{j\in\delta^-(i)} b_{ji}^{dk}\right) = \begin{cases} 1 & \text{if } i=(z_0,1) \\ -1 & \text{if } i=(z_0,T) \\ 0 & \text{otherwise} \end{cases} \quad \forall d\in D, \forall i\in N \tag{2}$$

$$u_{ij}^d + \sum_{k\in K} b_{ij}^{dk} \le 1 \quad \forall d\in D, \forall (i,j)\in A \tag{3}$$

$$b_{ij}^{dk} \le y_{ij}^k \quad d\in D, k\in K, (i,j)\in A \tag{4}$$

$$\sum_{d\in D} b_{ij}^{dk} \le Q_k^{\text{droid}} \cdot y_{ij}^k \tag{5}$$

$$\sum_{k\in K}\sum_{j\in\delta^+((z_0,1))} x_{(z_0,1),j}^{pk} + \sum_{k\in K}\sum_{d\in D} \alpha_{(z_0,1)}^{pkd} = 1 \quad \forall p\in P \tag{6a}$$

$$\alpha_{(z_0,1)}^{pkd} \le w_k \quad \forall p\in P, \forall k\in K, \forall d\in D \tag{6b}$$

$$\sum_{j\in\in\delta^+(i)} x_{ij}^{pk} - \sum_{j\in\in\delta^-(i)} x_{ji}^{pk} = -s_i^{pk} - \sum_{d\in D}\alpha_i^{pkd} + \sum_{d\in D}\beta_i^{pdk} \quad \forall p\in P, \forall k\in K, \forall i\in N\backslash\{(z_0,1)\} \tag{7}$$

$$\sum_{j\in\in\delta^+(i)} \xi_{ij}^{pd} - \sum_{j\in\in\delta^-(i)} \xi_{ji}^{pd} = -\sigma_i^{pd} + \sum_{k\in K} \alpha_i^{pkd} - \sum_{k\in K} \beta_i^{pdk} \quad \forall p\in P, \forall d\in D, \forall i\in N \tag{8}$$

$$\sum_{k\in K}\sum_{i\in N_p} s_i^{pk} + \sum_{d\in D}\sum_{i\in N_p} \sigma_i^{dp} = 1 \quad \forall p\in P \tag{9}$$

$$s_i^{pk} \leq \sum_{j\in\delta^+(i)} y_{ij}^k \quad \forall p\in P, \forall k\in K, \forall i\in N_p \tag{10a}$$

$$\sigma_i^{pd} \leq \sum_{j\in\delta^+(i)} u_{ij}^d \quad \forall p\in P, \forall d\in D, \forall i\in N_p \tag{10b}$$

$$x_{ij}^{pk} \leq y_{ij}^k \quad \forall p\in P, \forall k\in K, \forall (i,j)\in A \tag{11a}$$

$$\xi_{ij}^{pd} \leq u_{ij}^d \quad \forall p\in P, \forall d\in D, \forall (i,j)\in A \tag{11b}$$

$$\sum_{p\in P} x_{ij}^{pk} \leq Q_k^{\text{truck}} \cdot y_{ij}^k \quad \forall k\in K, \forall (i,j)\in A \tag{12}$$

$$\sum_{p\in P} \xi_{ij}^{pd} \leq Q_d^{\text{droid-parcel}} \cdot u_{ij}^d \quad \forall d\in D, \forall (i,j)\in A \tag{13}$$

$$\sum_{(i,j)\in A} e_{ij}^{\text{droid}} \cdot u_{ij}^d + \sum_{i\in N}\sum_{p\in P} \sigma_i^{pd} \leq R_d \quad \forall d\in D \tag{14}$$

$$\sum_{p:(z,t)\in N_p} \sigma_{(z,t)}^{pd} \leq S^{\max} \quad \forall d\in D, \forall (z,t)\in N \tag{15}$$

$$\sum_{j\in\delta^-(i=(z,T))} x_{j,(z,T)}^{pk} = 0 \quad \forall p\in P, \forall k\in K, \forall z\in Z \tag{16a}$$

$$\sum_{j\in\delta^-(i=(z,T))} \xi_{j,(z,T)}^{pd} = 0 \quad \forall p\in P, \forall k\in K, \forall z\in Z \tag{16b}$$

$$\alpha_i^{pkd} \leq \sum_{j\in\delta^+(i)} y_{ij}^k \quad \forall p\in P, \forall k\in K, \forall d\in D, \forall i\in N \tag{17a}$$

$$\alpha_i^{pkd} \leq \sum_{j\in\delta^+(i)} u_{ij}^d \quad \forall p\in P, \forall k\in K, \forall d\in D, \forall i\in N \tag{17b}$$

$$\beta_i^{pdk} \leq \sum_{j\in\delta^+(i)} u_{ij}^d \quad \forall p\in P, \forall k\in K, \forall d\in D, \forall i\in N \tag{18a}$$

$$\beta_i^{pdk} \leq \sum_{j\in\delta^+(i)} y_{ij}^k \quad \forall p\in P, \forall k\in K, \forall d\in D, \forall i\in N \tag{18b}$$

Explanation of Constraints

- Constraint 1 is the flow conservation for the trucks.
- Constraint 2 is the flow conservation constraint for the droids. This means that at each node i the droid either continuous operating normally, board/ carried by a truck or disembarks from a truck.
- Constraint 3 means that a droid can only in one state (independent operation or carried by truck).
- Constraint 4 states that a droid can only be carried by a truck if that uses the same arc.

- Constraint 5 is the droid carrying capacity. This means that the total number of droids carried by truck k on arc (i, j) cannot exceed its droid capacity Q_k^{droid}.

- Constraint 6a ensures that each delivery parcel originates at the depot exactly once at the beginning of the planning horizon (time period 1). Each parcel must follow one of two mutually exclusive paths. The first term $\sum_{k \in K} \sum_{j \in \delta^+((z_0,1))} x^{pk}_{(z_0,1),j}$ says that parcel p leaves the depot by some outgoing arc on a truck k with the truck directly carring the parcel away from the depot. The second term $\sum_{k \in K} \sum_{d \in D} \alpha^{pkd}_{(z_0,1)}$ says that parcel p is transffered from a truck k to a droid d at the depot. Because delivery parcels are available at the depot in period 1 and may be loaded and transferred at the depot in period 1 without consuming a time period this constraint models that instantaneous depot transfers where a truck "picks up" the parcel and immediately hands it to a droid. Direct droid-pickup from the depot-wharehouse are not allowed because only rucks have the capability to load from the depot and droids receive parcels only through transfers from trucks. Constraint 6b ensures that a parcel can only be transferred from truck k k to droid d at the depot if truck if truck k is actualy deployed.

- Constraint 7 is the parcel flow conservation constraint for the truck. It maintains flow conservation for each parcel on each truck at all nodes except the depot start ($\forall i \in N \backslash \{(z_0, 1)\}$). The net outlow then becomes:
  - Negative if the truck delivers the parcel ($-s_i^{pk}$)
  - Negative if the truck transfers the parcel to a droid ($-\sum_{d \in D} \alpha_i^{pkd}$)
  - Positive if the truck recieves a parcel from a droid ($+\sum_{d \in D} \beta_i^{pdk}$)
  - 0 otherwise if the parcel flows trough

- Constraint 8 maintains flow conservation for each parcel on each droid at all nodes. Net outflow equals:
  - Negative if the droid delivers the parcel ($-\sigma_i^{pd}$)
  - Positive if the droid recieves the parcel from a truck $+\sum_{k \in K} \alpha_i^{pkd}$
  - Negative if the droid transfers a parcel to a truck $-\sum_{k \in K} \beta_i^{pdk}$
  - 0 otherwise if the parcel flows trough

  It must be stated that in this constraint the droids can only pick up parcels from trucks via transfers, not from the depot directly.

- Constraint 9 states that each service must only be serviced once.

- The constraints 10a and 10b states that a truck can only serve a request if it is at that location and that a droid can only serve a request if it is operating independently at that location.

- The constraint 11a and 11b link the parcel flow with the vehicle usages for the trucks and droids respectively.

- Constraint 12 is the truck parcel capacity. It states that the total parcels carried by truck k on arc (i, j) cannot exceed its parcel capacity Q_k^{truck}.

- Constraint 13 is the droid parcel capacity. It states that the total parcels carried by droid d on arc (i, j) cannot exceed its parcel capacity $Q_d^{\text{droid-parcel}}$.

- Constraint 14 states that the total energy consumed by droid d over the horizon must not exceed its energy budget R_d. In this constraint, the first term $\sum_{(i,j) \in A} e_{ij}^{\text{droid}} \cdot u_{ij}^d$ is the movement energy and the second term $\sum_{i \in N} \sum_{p \in P} \sigma_i^{pd}$ is the energy for delivery systems with 1 unit of energy per service.

- This constraint 15 states that a droid d can serve at most $S^{\max}$ requests per zone z during period t.

- Constraint 16a and 16b states that all parcel p must be delivered at the end of the horizon. All deliveries must be complete by time $T-1$. So for both droids as truck no parcels can arrive at time T.

- Constraints 17a and 17b are for the fact that a transfer from truck to droid requires a co-location. A parcel can only be transferred from truck k to droid d at node i if both vehicles are present at that node. Constraint 17a ensures that the truck is there and constraint 17b ensures that the droid is operating independent there. So if parcel p is transferred from truck k to droid d at node i (i.e. $\alpha_i^{pkd} = 1$) then truck k must be present at node i. Also, droid d must also be present and operating independently (forced by constraint 17b) at node i. A droid cannot be carried because due to constraint 3: the decision variable u_{ij}^d represents independent operation. If a droid is being carried ($b_{ij}^{dk} = 1$) then $u_{ij}^d = 0$ and so a carried droid cannot recieve parcels.

- Constraints 18a and 18b are for the fact that a transfer from a droid to a truck requires a co-location. A parcel can only be transferred from droid d to truck k at node i if both vehicles are at that node. Constraint 18a ensures that the droid is there and constraint 18b ensures that the truck is there. This is similair to constraints 17a and 17b, but there is need of four of these co-location constraints due to the fact there are 2 types of transfer variables $\alpha_i^{pkd} \in \{0,1\}$ and $\beta_i^{pkd} \in \{0,1\}$. Each transfer direction has its own variable and needs its own pair of co-location constraints. So that means:

  - Transfer type: truck $k \rightarrow$ droid d has transfer variable $\alpha_i^{pkd} \in \{0,1\}$ and requiers truck present (constraint 17a) and droid present (constraint 17b).
  - Transfer type: droid $d \rightarrow$ truck k has transfer variable $\beta_i^{pkd} \in \{0,1\}$ and requiers droid present (constraint 18a) and truck present (constraint 18b).

  The presence is checked by looking at the outgoing arcs $\delta^+(i)$ because in a time-space network being at a node means there are arcs leaving that node.

Sets and Indices

Set	Description	Index
Z	Set of physical zones	z
$Z'(z) \subseteq Z$	Set zones adjacent to zone z	z'
$\mathcal{T} = \{1, ..., T\}$	Set of time steps	t
$N = Z \times \mathcal{T}$	Time-space nodes	$i = (z, t)$
$A \subseteq N \times N$	Time-space arcs: $A = \Big\{\big((z,t),(z',t+1)\big) : z' \in Z'(z) \cup \{z\}, z \in Z, t \in \mathcal{T}/\{T\}\Big\}$	$a = (i, j)$
$A^{move} \subseteq A$	Movement arcs: $A^{move} = \Big\{\big((z,t),(z',t+1)\big) \in A : z' \in Z'(z), z \in Z, t \in \mathcal{T}/\{T\}\Big\}$	$a = (i, j)$
$A^{stay} \subseteq A$	Waiting arcs: $A^{stay} = \Big\{\big((z,t),(z,t+1)\big) \in A, z \in Z, t \in \mathcal{T}/\{T\}\Big\}$	$a = (i, j)$
$A(\bar{t}) \subseteq A$	Time-space arcs until timestep $\bar{t}$: $A(\bar{t}) = \Big\{\big((z,t),(z',t+1)\big) \in A : z' \in Z'(z) \cup \{z\}, z \in Z, t \in \{1, 2, ..., \bar{t}-1\}\Big\}$	$a = (i, j)$
$\delta^+(i) \subseteq A$	Outgoing arcs from node $i \in N$: $\delta^+\big((z,t)\big) = \Big\{\big((z,t),(z',t+1)\big) \in A : z' \in Z'(z) \cup \{z\}\Big\}$	$a = (i, j)$
$\delta^-(i) \subseteq A$	Incoming arcs to node $i \in N$: $\delta^-\big((z,t)\big) = \Big\{\big((z',t-1),(z,t)\big) \in A : z' \in Z'(z) \cup \{z\}\Big\}$	$a = (i, j)$
K	Set of trucks	k
D	Set of droids	d
$\mathcal{R}$	Set of parcel delivery requests	r
$\mathcal{T}_r \subseteq \mathcal{T}$	Set of feasible delivery times for delivery request r: $\mathcal{T}_r = \{t \in \mathcal{T} : t \leq \bar{t}_r\}$	t

Parameters

Param.	Description	Remarks
z_0	Depot zone	
T	Final timestep	$\lvert\mathcal{T}\rvert = T$
c_a^t	Cost associated with arc a for trucks	$\forall a \in A$
c_{fixed}^t	Fixed cost for deploying a truck	
c_{energy}^d	Cost per used unit of energy	
Q_{droid}^t	Maximal droid capacity of a truck	
Q_{parcel}^t	Maximal parcel capacity of a truck	
Q_{parcel}^d	Maximal parcel capacity of a droid	
R	Energy budget for a droid	
e_a	Energy cost for arc a	$e_a = 1$ if $a \in A^{move}$ $e_a = 0$ if $a \in A^{stay}$
$a_{stay}^+(i)$	Stay arc going out of node i	$a_{stay}^+(i) \in \delta^+(i) \cap A^{stay}$
$a_{stay}^-(i)$	Stay arc coming into node i	$a_{stay}^-(i) \in \delta^-(i) \cap A^{stay}$
S^{max}	Maximum number of requests a droid can serve during a time step	
$\bar{z}_r$	Destination zone of delivery request r	$\bar{z}_r \in Z$, $r \in \mathcal{R}$
$\bar{t}_r$	Delivery deadline of delivery request r	$\bar{t}_r \in \mathcal{T}$, $r \in \mathcal{R}$

Decision variables

Type	Var.	Description	
Bin.	$x_{k,a}$	1 if truck k traverses arc a 0 otherwise	$\forall k \in K$, $\forall a \in A$
Bin.	y_k	1 if truck k is deployed 0 otherwise	$\forall k \in K$
Bin.	$u_{d,a}$	1 if droid d traverses arc a independently 0 otherwise	$\forall d \in D$, $\forall a \in A$
Bin.	$v_{d,k,a}$	1 if droid d traverses arc a carried by truck k 0 otherwise	$\forall d \in D$, $\forall k \in K$, $\forall a \in A$
Bin.	w_d	1 if droid d is deployed 0 otherwise	$\forall d \in D$
Bin.	$f_{r,k,a}$	1 if parcel r is riding truck k on arc a 0 otherwise	$\forall r \in \mathcal{R}$, $\forall k \in K$, $\forall a \in A$
Bin.	$g_{r,d,a}$	1 if parcel r is riding droid d on arc a 0 otherwise	$\forall r \in \mathcal{R}$, $\forall d \in D$, $\forall a \in A$
Bin.	$h_{r,a}$	1 if parcel r is riding stay arc a in zone z 0 otherwise	$\forall r \in \mathcal{R}$, $\forall a \in A^{stay}$
Bin.	$s^t_{r,k,t}$	1 if parcel r is serviced by truck k at timestep t 0 otherwise	$\forall r \in \mathcal{R}$, $\forall k \in K$, $\forall t \in \mathcal{T}_r$
Bin.	$s^d_{r,d,t}$	1 if parcel r is serviced by droid d at timestep t 0 otherwise	$\forall r \in \mathcal{R}$, $\forall d \in D$, $\forall t \in \mathcal{T}_r$

Objective and constrains

Objective function:

$$min \sum_{k\in K}(c^t_{fixed}\cdot y_k + \sum_{a\in A} c^t_a \cdot x_{k,a}) + c^d_{energy}\sum_{d\in D}(\sum_{a\in A} e_a u_{d,a} + \sum_{r\in\mathcal{R}}\sum_{t\in\mathcal{T}_r} s^d_{r,d,t}) \tag{1}$$

S.t.:

Truck movement

$$\sum_{a\in\delta^+(i)} x_{k,a} - \sum_{a\in\delta^-(i)} x_{k,a} = 0, \quad \forall i \in N/\{(z_0,1),(z_0,T)\}, \forall k \in K \tag{2}$$

$$\sum_{a\in\delta^+((z_0,1))} x_{k,a} = y_k, \quad \forall k \in K \tag{3}$$

$$\sum_{a\in\delta^-((z_0,T))} x_{k,a} = y_k, \quad \forall k \in K \tag{4}$$

Droid movement

$$\sum_{a\in\delta^+(i)}(u_{d,a} + \sum_{k\in K} v_{d,k,a}) - \sum_{a\in\delta^-(i)}(u_{d,a} + \sum_{k\in K} v_{d,k,a}) = 0, \quad \forall i \in N/\{(z_0,1),(z_0,T)\}, \forall d \in D \tag{5}$$

$$\sum_{a\in\delta^+((z_0,1))}(u_{d,a} + \sum_{k\in K} v_{d,k,a}) = w_d, \quad \forall d \in D \tag{6}$$

$$\sum_{a\in\delta^-((z_0,T))}(u_{d,a} + \sum_{k\in K} v_{d,k,a}) = w_d, \quad \forall d \in D \tag{7}$$

$$\sum_{d\in D} v_{d,k,a} \le Q^t_{droid}\cdot x_{k,a}, \quad \forall k \in K, \forall a \in A \tag{8}$$

$$\sum_{a\in A} e_a u_{d,a} + \sum_{r\in\mathcal{R}}\sum_{t\in\mathcal{T}_r} s^d_{d,r,t} \le R w_d, \quad \forall d \in D \tag{9}$$

Parcel movement

$$\sum_{a\in\delta^+(i)}(\sum_{k\in K} f_{r,k,a} + \sum_{d\in D} g_{r,d,a}) + h_{r,a^+_{stay}(i)} - \sum_{a\in\delta^-(i)}(\sum_{k\in K} f_{r,k,a} + \sum_{d\in D} g_{r,d,a}) - h_{r,a^-_{stay}(i)} = 0, \quad \forall i \in (Z\times\mathcal{T}_r)/\{(z_0,1),(\overline{z}_r,t) : t\in\mathcal{T}_r\}, \forall r \in \mathcal{R} \tag{10}$$

$$\sum_{a\in\delta^+((z_0,1))}(\sum_{k\in K} f_{r,k,a} + \sum_{d\in D} g_{r,d,a}) + h_{r,a^+_{stay}((z_0,1))} = 1, \quad \forall r \in \mathcal{R} \tag{11}$$

$$\sum_{a\in\delta^+((\overline{z}_r,t))}(\sum_{k\in K} f_{r,k,a} + \sum_{d\in D} g_{r,d,a}) - \sum_{a\in\delta^-((\overline{z}_r,t))}(\sum_{k\in K} f_{r,k,a} + \sum_{d\in D} g_{r,d,a}) = -(\sum_{k\in K} s^t_{r,k,t} + \sum_{d\in D} s^d_{r,d,t}), \quad \forall r \in \mathcal{R}, \forall t \in \mathcal{T}_r \tag{12}$$

$$\sum_{r\in\mathcal{R}} f_{r,k,a} \le Q^t_{parcel}\cdot x_{k,a}, \quad \forall k \in K, \forall a \in A \tag{13}$$

$$\sum_{r\in\mathcal{R}} g_{r,d,a} \le Q^d_{parcel}\cdot u_{d,a}, \quad \forall d \in D, \forall a \in A \tag{14}$$

$$\sum_{a\in A/A(\overline{t}_r)} f_{r,k,a} = 0, \quad \forall r \in \mathcal{R}, \forall k \in K \tag{15a}$$

$$\sum_{a\in A/A(\overline{t}_r)} g_{r,d,a} = 0, \quad \forall r \in \mathcal{R}, \forall d \in D \tag{15b}$$

$$h_{r,a} = 0, \quad \forall r \in \mathcal{R}, \forall a \in A^{stay}/\{((z_0,t),(z_0,t+1)) : t \in \mathcal{T}/\{T\}\} \tag{16}$$

Parcel Delivery

$$\sum_{t\in\mathcal{T}_r}\sum_{k\in K} s^t_{r,k,t} + \sum_{t\in\mathcal{T}_r}\sum_{d\in D} s^d_{r,d,t} = 1, \quad \forall r \in \mathcal{R} \tag{17}$$

$$s^t_{r,k,t} \leq f_{r,k,a^-_{stay}((\overline{z}_r,t))}, \quad \forall k \in K, t \in \mathcal{T}_r, \forall r \in \mathcal{R} \tag{18}$$

$$\sum_{r\in\mathcal{R}:t\leq\overline{t}_r} s^t_{r,k,t} \leq 1, \quad \forall k \in K, t \in \mathcal{T} \tag{19}$$

$$s^d_{r,d,t} \leq g_{r,d,a^-_{stay}((\overline{z}_r,t))}, \quad \forall d \in D, t \in \mathcal{T}_r, \forall r \in \mathcal{R} \tag{20}$$

$$\sum_{r\in\mathcal{R}:t\leq\overline{t}_r} s^d_{r,d,t} \leq S^{max}, \quad \forall d \in D, t \in \mathcal{T} \tag{21}$$

Parcel transfer limitations

$$\sum_{a\in\delta^-(i)} f_{r,k,a} + \sum_{a\in\delta^+(i)} f_{r,k',a} \leq 1, \quad \forall r \in \mathcal{R}, \forall k, k' \neq k \in K, \forall i \in N \tag{22}$$

$$\sum_{a\in\delta^-(i)} g_{r,d,a} + \sum_{a\in\delta^+(i)} g_{r,d',a} \leq 1, \quad \forall r \in \mathcal{R}, \forall d, d' \neq d \in D, \forall i \in N \tag{23}$$

Explanation

1. Objective function; minimize total cost
2. The number of flow arcs for truck k into node i should be equal to number of flow arcs out of node i.
3. If truck k is deployed, exactly 1 flow arc leaves the depot zone z_0 at $t = 1$ for truck k; together with constraint 2, this constraint enforces that a truck visits any node max. once.
7. If truck k is deployed, exactly 1 flow arc enters the depot zone z_0 at $t = T$ for truck k.
5. The number of flow arcs for droid d into node i should be equal to number of flow arcs out of node i.
6. If droid d is deployed, exactly 1 flow arc leaves the depot zone z_0 at $t = 1$ for droid d; together with constraint 5, this constraint enforces that a droid visits any node max. once.
7. If droid d is deployed, exactly 1 flow arc enters the depot zone z_0 at $t = T$ for droid d.
8. Enforces no more than Q^t_{droid} droids ride truck k on the same arc.
9. Limits the amount of energy used by droid d.
10. The number of flow arcs for delivery request r into node i should be equal to number of flow arcs out of node i.
11. Exactly 1 flow arc leaves the depot zone z_0 at $t = 1$ delivery request r; together with constraint 10, this constraint enforces that a parcel visits any node max. once.
12. If parcel r is serviced at time step t, exactly 1 flow arc enters node $(\overline{z}_r, t)$, and none leave that node.
13. Enforces no more than Q^t_{parcel} parcels ride truck k on the same arc.
14. Enforces no more than Q^d_{parcel} parcels ride droid d on the same arc, and only when droid d traverses the arc independently.
15. No parcel flow for request r after $\overline{t}_r$.
17. Any parcel should be delivered by max. one vehicle.
18. Parcel r can only be delivered by truck k at timestep t if the truck arrives at node $(\overline{z}_r, t)$.
19. Max. 1 delivery per truck per timestep.
20. Parcel r can only be delivered by droid d at timestep t if the droid arrives at node $(\overline{z}_r, t)$.
21. Max. S^{max} deliveries per droid per timestep.
23. No droid-droid transfers allowed.

Sets and Indices

Set	Description	Index
Z	Set of physical zones	z
$Z'(z) \subseteq Z$	Set zones adjacent to zone z	z'
$\mathcal{T} = \{1, ..., T\}$	Set of time steps	t
$N = Z \times \mathcal{T}$	Time-space nodes	$i = (z, t)$
$A \subseteq N \times N$	Time-space arcs: $A = \Big\{\big((z,t),(z',t+1)\big) : z' \in Z'(z) \cup \{z\}, z \in Z, t \in \mathcal{T}/\{T\}\Big\}$	$a = (i, j)$
$A^{move} \subseteq A$	Movement arcs: $A^{move} = \Big\{\big((z,t),(z',t+1)\big) \in A : z' \in Z'(z), z \in Z, t \in \mathcal{T}/\{T\}\Big\}$	$a = (i, j)$
$A^{stay} \subseteq A$	Waiting arcs: $A^{stay} = \Big\{\big((z,t),(z,t+1)\big) \in A, z \in Z, t \in \mathcal{T}/\{T\}\Big\}$	$a = (i, j)$
$A(\bar{t}) \subseteq A$	Time-space arcs until timestep $\bar{t}$: $A(\bar{t}) = \Big\{\big((z,t),(z',t+1)\big) \in A : z' \in Z'(z) \cup \{z\}, z \in Z, t \in \{1, 2, ..., \bar{t}-1\}\Big\}$	$a = (i, j)$
$\delta^+(i) \subseteq A$	Outgoing arcs from node $i \in N$: $\delta^+\big((z,t)\big) = \Big\{\big((z,t),(z',t+1)\big) \in A : z' \in Z'(z) \cup \{z\}\Big\}$	$a = (i, j)$
$\delta^-(i) \subseteq A$	Incoming arcs to node $i \in N$: $\delta^-\big((z,t)\big) = \Big\{\big((z',t-1),(z,t)\big) \in A : z' \in Z'(z) \cup \{z\}\Big\}$	$a = (i, j)$
K	Set of trucks	k
D	Set of droids	d
$\mathcal{R}$	Set of parcel delivery requests	r
$\mathcal{T}_r \subseteq \mathcal{T}$	Set of feasible delivery times for delivery request r: $\mathcal{T}_r = \{t \in \mathcal{T} : t \leq \bar{t}_r\}$	t

Parameters

Param.	Description	Remarks
z_0	Depot zone	
T	Final timestep	$\lvert\mathcal{T}\rvert = T$
c_a^t	Cost associated with arc a for trucks	$\forall a \in A$
c_{fixed}^t	Fixed cost for deploying a truck	
c_{energy}^d	Cost per used unit of energy	
Q_{droid}^t	Maximal droid capacity of a truck	
Q_{parcel}^t	Maximal parcel capacity of a truck	
Q_{parcel}^d	Maximal parcel capacity of a droid	
R	Energy budget for a droid	
e_a	Energy cost for arc a	$e_a = 1$ if $a \in A^{move}$ $e_a = 0$ if $a \in A^{stay}$
$a_{stay}^+(i)$	Stay arc going out of node i	$a_{stay}^+(i) \in \delta^+(i) \cap A^{stay}$
$a_{stay}^-(i)$	Stay arc coming into node i	$a_{stay}^-(i) \in \delta^-(i) \cap A^{stay}$
S^{max}	Maximum number of requests a droid can serve during a time step	
$\bar{z}_r$	Destination zone of delivery request r	$\bar{z}_r \in Z$, $r \in \mathcal{R}$
$\bar{t}_r$	Delivery deadline of delivery request r	$\bar{t}_r \in \mathcal{T}$, $r \in \mathcal{R}$

Decision variables

Type	Var.	Description	
Bin.	$x_{k,a}$	1 if truck k traverses arc a 0 otherwise	$\forall k \in K$, $\forall a \in A$
Bin.	y_k	1 if truck k is deployed 0 otherwise	$\forall k \in K$
Bin.	$u_{d,a}$	1 if droid d traverses arc a independently 0 otherwise	$\forall d \in D$, $\forall a \in A$
Bin.	$v_{d,k,a}$	1 if droid d traverses arc a carried by truck k 0 otherwise	$\forall d \in D$, $\forall k \in K$, $\forall a \in A$
Bin.	w_d	1 if droid d is deployed 0 otherwise	$\forall d \in D$
Bin.	$f_{r,k,a}$	1 if parcel r is riding truck k on arc a 0 otherwise	$\forall r \in \mathcal{R}$, $\forall k \in K$, $\forall a \in A$
Bin.	$g_{r,d,a}$	1 if parcel r is riding droid d on arc a 0 otherwise	$\forall r \in \mathcal{R}$, $\forall d \in D$, $\forall a \in A$
Bin.	$h_{r,a}$	1 if parcel r is riding stay arc a in zone z 0 otherwise	$\forall r \in \mathcal{R}$, $\forall a \in A^{stay}$
Bin.	$s^t_{r,k,t}$	1 if parcel r is serviced by truck k at timestep t 0 otherwise	$\forall r \in \mathcal{R}$, $\forall k \in K$, $\forall t \in \mathcal{T}_r$
Bin.	$s^d_{r,d,t}$	1 if parcel r is serviced by droid d at timestep t 0 otherwise	$\forall r \in \mathcal{R}$, $\forall d \in D$, $\forall t \in \mathcal{T}_r$

Objective and constrains

Objective function:

$$min \sum_{k\in K}(c^t_{fixed}\cdot y_k + \sum_{a\in A} c^t_a \cdot x_{k,a}) + c^d_{energy}\sum_{d\in D}(\sum_{a\in A} e_a u_{d,a} + \sum_{r\in\mathcal{R}}\sum_{t\in\mathcal{T}_r} s^d_{r,d,t}) \tag{1}$$

S.t.:

Truck movement

$$\sum_{a\in\delta^+(i)} x_{k,a} - \sum_{a\in\delta^-(i)} x_{k,a} = 0, \quad \forall i \in N/\{(z_0,1),(z_0,T)\}, \forall k \in K \tag{2}$$

$$\sum_{a\in\delta^+((z_0,1))} x_{k,a} = y_k, \quad \forall k \in K \tag{3}$$

$$\sum_{a\in\delta^-((z_0,T))} x_{k,a} = y_k, \quad \forall k \in K \tag{4}$$

Droid movement

$$\sum_{a\in\delta^+(i)}(u_{d,a} + \sum_{k\in K} v_{d,k,a}) - \sum_{a\in\delta^-(i)}(u_{d,a} + \sum_{k\in K} v_{d,k,a}) = 0, \quad \forall i \in N/\{(z_0,1),(z_0,T)\}, \forall d \in D \tag{5}$$

$$\sum_{a\in\delta^+((z_0,1))}(u_{d,a} + \sum_{k\in K} v_{d,k,a}) = w_d, \quad \forall d \in D \tag{6}$$

$$\sum_{a\in\delta^-((z_0,T))}(u_{d,a} + \sum_{k\in K} v_{d,k,a}) = w_d, \quad \forall d \in D \tag{7}$$

$$\sum_{d\in D} v_{d,k,a} \le Q^t_{droid}\cdot x_{k,a}, \quad \forall k \in K, \forall a \in A \tag{8}$$

$$\sum_{a\in A} e_a u_{d,a} + \sum_{r\in\mathcal{R}}\sum_{t\in\mathcal{T}_r} s^d_{d,r,t} \le R w_d, \quad \forall d \in D \tag{9}$$

Parcel movement

$$\sum_{a\in\delta^+(i)}(\sum_{k\in K} f_{r,k,a} + \sum_{d\in D} g_{r,d,a}) + h_{r,a^+_{stay}(i)} - \sum_{a\in\delta^-(i)}(\sum_{k\in K} f_{r,k,a} + \sum_{d\in D} g_{r,d,a}) - h_{r,a^-_{stay}(i)} = 0, \quad \forall i \in (Z\times\mathcal{T}_r)/\{(z_0,1),(\overline{z}_r,t) : t\in\mathcal{T}_r\}, \forall r \in \mathcal{R} \tag{10}$$

$$\sum_{a\in\delta^+((z_0,1))}(\sum_{k\in K} f_{r,k,a} + \sum_{d\in D} g_{r,d,a}) + h_{r,a^+_{stay}((z_0,1))} = 1, \quad \forall r \in \mathcal{R} \tag{11}$$

$$\sum_{a\in\delta^+((\overline{z}_r,t))}(\sum_{k\in K} f_{r,k,a} + \sum_{d\in D} g_{r,d,a}) - \sum_{a\in\delta^-((\overline{z}_r,t))}(\sum_{k\in K} f_{r,k,a} + \sum_{d\in D} g_{r,d,a}) = -(\sum_{k\in K} s^t_{r,k,t} + \sum_{d\in D} s^d_{r,d,t}), \quad \forall r \in \mathcal{R}, \forall t \in \mathcal{T}_r \tag{12}$$

$$\sum_{r\in\mathcal{R}} f_{r,k,a} \le Q^t_{parcel}\cdot x_{k,a}, \quad \forall k \in K, \forall a \in A \tag{13}$$

$$\sum_{r\in\mathcal{R}} g_{r,d,a} \le Q^d_{parcel}\cdot u_{d,a}, \quad \forall d \in D, \forall a \in A \tag{14}$$

$$\sum_{a\in A/A(\overline{t}_r)} f_{r,k,a} = 0, \quad \forall r \in \mathcal{R}, \forall k \in K \tag{15a}$$

$$\sum_{a\in A/A(\overline{t}_r)} g_{r,d,a} = 0, \quad \forall r \in \mathcal{R}, \forall d \in D \tag{15b}$$

$$h_{r,a} = 0, \quad \forall r \in \mathcal{R}, \forall a \in A^{stay}/\{((z_0,t),(z_0,t+1)) : t \in \mathcal{T}/\{T\}\} \tag{16}$$

Parcel Delivery

$$\sum_{t\in\mathcal{T}_r}\sum_{k\in K} s^t_{r,k,t} + \sum_{t\in\mathcal{T}_r}\sum_{d\in D} s^d_{r,d,t} = 1, \quad \forall r \in \mathcal{R} \tag{17}$$

$$s^t_{r,k,t} \leq f_{r,k,a^-_{stay}((\overline{z}_r,t))}, \quad \forall k \in K, t \in \mathcal{T}_r, \forall r \in \mathcal{R} \tag{18}$$

$$\sum_{r\in\mathcal{R}:t\leq\overline{t}_r} s^t_{r,k,t} \leq 1, \quad \forall k \in K, t \in \mathcal{T} \tag{19}$$

$$s^d_{r,d,t} \leq g_{r,d,a^-_{stay}((\overline{z}_r,t))}, \quad \forall d \in D, t \in \mathcal{T}_r, \forall r \in \mathcal{R} \tag{20}$$

$$\sum_{r\in\mathcal{R}:t\leq\overline{t}_r} s^d_{r,d,t} \leq S^{max}, \quad \forall d \in D, t \in \mathcal{T} \tag{21}$$

Parcel transfer limitations

$$\sum_{a\in\delta^-(i)} f_{r,k,a} + \sum_{a\in\delta^+(i)} f_{r,k',a} \leq 1, \quad \forall r \in \mathcal{R}, \forall k, k' \neq k \in K, \forall i \in N \tag{22}$$

$$\sum_{a\in\delta^-(i)} g_{r,d,a} + \sum_{a\in\delta^+(i)} g_{r,d',a} \leq 1, \quad \forall r \in \mathcal{R}, \forall d, d' \neq d \in D, \forall i \in N \tag{23}$$

Explanation

1. Objective function; minimize total cost
2. The number of flow arcs for truck k into node i should be equal to number of flow arcs out of node i.
3. If truck k is deployed, exactly 1 flow arc leaves the depot zone z_0 at $t = 1$ for truck k; together with constraint 2, this constraint enforces that a truck visits any node max. once.
7. If truck k is deployed, exactly 1 flow arc enters the depot zone z_0 at $t = T$ for truck k.
5. The number of flow arcs for droid d into node i should be equal to number of flow arcs out of node i.
6. If droid d is deployed, exactly 1 flow arc leaves the depot zone z_0 at $t = 1$ for droid d; together with constraint 5, this constraint enforces that a droid visits any node max. once.
7. If droid d is deployed, exactly 1 flow arc enters the depot zone z_0 at $t = T$ for droid d.
8. Enforces no more than Q^t_{droid} droids ride truck k on the same arc.
9. Limits the amount of energy used by droid d.
10. The number of flow arcs for delivery request r into node i should be equal to number of flow arcs out of node i.
11. Exactly 1 flow arc leaves the depot zone z_0 at $t = 1$ delivery request r; together with constraint 10, this constraint enforces that a parcel visits any node max. once.
12. If parcel r is serviced at time step t, exactly 1 flow arc enters node $(\overline{z}_r, t)$, and none leave that node.
13. Enforces no more than Q^t_{parcel} parcels ride truck k on the same arc.
14. Enforces no more than Q^d_{parcel} parcels ride droid d on the same arc, and only when droid d traverses the arc independently.
15. No parcel flow for request r after $\overline{t}_r$.
17. Any parcel should be delivered by max. one vehicle.
18. Parcel r can only be delivered by truck k at timestep t if the truck arrives at node $(\overline{z}_r, t)$.
19. Max. 1 delivery per truck per timestep.
20. Parcel r can only be delivered by droid d at timestep t if the droid arrives at node $(\overline{z}_r, t)$.
21. Max. S^{max} deliveries per droid per timestep.
23. No droid-droid transfers allowed.

Sets and Indices

Set	Description	Index
Z	Set of physical zones	z
$Z'(z) \subseteq Z$	Set zones adjacent to zone z	z'
$\mathcal{T} = \{1, ..., T\}$	Set of time steps	t
$N = Z \times \mathcal{T}$	Time-space nodes	$i = (z, t)$
$A \subseteq N \times N$	Time-space arcs: $A = \Big\{\big((z,t),(z',t+1)\big) : z' \in Z'(z) \cup \{z\}, z \in Z, t \in \{1, \ldots, T\}\Big\}$	$a = (i, j)$
$A^{move} \subseteq A$	Movement arcs: $A^{move} = \Big\{\big((z,t),(z',t+1)\big) \in A : z' \in Z'(z)\Big\}$	$a = (i, j)$
$A^{stay} \subseteq A$	Waiting arcs: $A^{stay} = \Big\{\big((z,t),(z,t+1)\big) \in A\Big\}$	$a = (i, j)$
$A(\bar{t}) \subseteq A$	Time-space arcs until time step $\bar{t}$: $A(\bar{t}) = \Big\{\big((z,t),(z',t+1)\big) \in A : z' \in Z'(z) \cup \{z\}, z \in Z, t \in \{1, 2, ..., \bar{t}\}\Big\}$	$a = (i, j)$
$\delta^+(i) \subseteq A$	Outgoing arcs from node $i \in N$: $\delta^+(i) = \{(i, j) \in A\}$	$a = (i, j)$
$\delta^-(i) \subseteq A$	Incoming arcs to node $i \in N$: $\delta^-(i) = \{(j, i) \in A\}$	$a = (i, j)$
K	Set of trucks	k
D	Set of droids	d
$\mathcal{R}$	Set of parcel delivery requests	r
$\mathcal{T}_r \subseteq \mathcal{T}$	Set of feasible delivery service times for delivery request r: $\mathcal{T}_r = \{t \in \mathcal{T} : t \leq \bar{t}_r \ \wedge \ t \leq T - 1\}$	t
$\mathcal{P}$	Set of pickup requests	p
$\mathcal{T}_p \subseteq \mathcal{T}$	Set of feasible pickup service times for pickup request p: $\mathcal{T}_p = \{t \in \mathcal{T} : t \geq \bar{t}_p\}$	t
$A^p(\bar{t}) \subseteq A$	Time-space arcs available for pickup p (departures after release): $A^p(\bar{t}) = \{((z,t),(z',t+1)) \in A : t \geq \bar{t}\}$	$a = (i, j)$

Parameters

Param.	Description	Remarks
z_0	Depot zone	
T	Final timestep	$\lvert\mathcal{T}\rvert = T$
c_a^t	Cost associated with arc a for trucks	$\forall a \in A$
c_{fixed}^t	Fixed cost for deploying a truck	
c_{energy}^d	Cost per used unit of energy	
Q_{droid}^t	Maximal droid capacity of a truck	
Q_{parcel}^t	Maximal parcel capacity of a truck	
Q_{parcel}^d	Maximal parcel capacity of a droid	
R	Energy budget for a droid	
e_a	Energy cost for arc a	$e_a = 1$ if $a \in A^{move}$ $e_a = 0$ if $a \in A^{stay}$
$a_{stay}^+(i)$	Stay arc going out of node i	$a_{stay}^+(i) \in \delta^+(i) \cap A^{stay}$
$a_{stay}^-(i)$	Stay arc coming into node i	$a_{stay}^-(i) \in \delta^-(i) \cap A^{stay}$
S^{max}	Maximum number of requests a droid can serve during a time step	
$\bar{z}_r$	Destination zone of delivery request r	$\bar{z}_r \in Z,\ r \in \mathcal{R}$
$\bar{t}_r$	Delivery deadline of delivery request r	$\bar{t}_r \in \mathcal{T},\ r \in \mathcal{R}$
M	Big-M constant for energy reset constraints	$M = 2R$
$\bar{z}_p$	Pickup zone (origin) of pickup request p	$\bar{z}_p \in Z,\ p \in P$
$\bar{t}_p$	Release time of pickup request p	$\bar{t}_p \in \mathcal{T},\ p \in P$

Decision Variables

Type	Var.	Description	
Bin.	$x_{k,a}$	1 if truck k traverses arc a 0 otherwise	$\forall k \in K$, $\forall a \in A$
Bin.	y_k	1 if truck k is deployed 0 otherwise	$\forall k \in K$
Bin.	$u_{d,a}$	1 if droid d traverses arc a independently 0 otherwise	$\forall d \in D$, $\forall a \in A$
Bin.	$v_{d,k,a}$	1 if droid d traverses arc a carried by truck k 0 otherwise	$\forall d \in D$, $\forall k \in K$, $\forall a \in A$
Bin.	w_d	1 if droid d is deployed 0 otherwise	$\forall d \in D$
Bin.	$f^r_{r,k,a}$	1 if parcel r is riding truck k on arc a 0 otherwise	$\forall r \in \mathcal{R}$, $\forall k \in K$, $\forall a \in A$
Bin.	$g^r_{r,d,a}$	1 if parcel r is riding droid d on arc a 0 otherwise	$\forall r \in \mathcal{R}$, $\forall d \in D$, $\forall a \in A$
Bin.	$h^r_{r,a}$	1 if parcel r is riding stay arc a in zone z 0 otherwise	$\forall r \in \mathcal{R}$, $\forall a \in A^{stay}$
Bin.	$s^{rt}_{r,k,t}$	1 if parcel r is serviced by truck k at timestep t 0 otherwise	$\forall r \in \mathcal{R}$, $\forall k \in K$, $\forall t \in \mathcal{T}_r$
Bin.	$s^{rt}_{r,d,t}$	1 if parcel r is serviced by droid d at timestep t 0 otherwise	$\forall r \in \mathcal{R}$, $\forall d \in D$, $\forall t \in \mathcal{T}_r$
Bin.	$f^p_{p,k,a}$	1 if pickup parcel p is riding truck k on arc a 0 otherwise	$\forall p \in P$, $\forall k \in K$, $\forall a \in A^p(\underline{t}_p)$
Bin.	$g^p_{p,d,a}$	1 if pickup parcel p is riding droid d on arc a 0 otherwise	$\forall p \in P$, $\forall d \in D$, $\forall a \in A^p(\underline{t}_p)$
Bin.	$h^p_{p,a}$	1 if pickup parcel p is stored at depot on stay arc a 0 otherwise	$\forall p \in P$, $\forall a = ((z_0, t), (z_0, t+1)) \in A^{stay}$
Bin.	$s^{pt}_{p,k,t}$	1 if pickup request p is served by truck k at timestep t 0 otherwise	$\forall p \in P$, $\forall k \in K$, $\forall t \in \mathcal{T}_p$
Bin.	$s^{pd}_{p,d,t}$	1 if pickup request p is served by droid d at timestep t 0 otherwise	$\forall p \in P$, $\forall d \in D$, $\forall t \in \mathcal{T}_p$
Bin.	$e^{reset}_{d,t}$	1 if droid d performs an energy reset at timestep t 0 otherwise	$\forall d \in D$, $\forall t \in \mathcal{T}$
Int.	$e^{counter}_{d,t}$	remaining available energy of droid d at timestep t, where $0 \leq e^{counter}_{d,t} \leq Rw_d$ and $e^{counter}_{d,1} = R$	$\forall d \in D$, $\forall t \in \mathcal{T}$

Objective and Constrains

Objective function:

$$\min \sum_{k\in K}\left(c^t_{fixed}\cdot y_k+\sum_{a\in A}c^t_a\cdot x_{k,a}\right)+c^d_{energy}\sum_{d\in D}\left(\sum_{a\in A}e_a u_{d,a}+\sum_{r\in\mathcal{R}}\sum_{t\in\mathcal{T}_r}s^{rd}_{r,d,t}+\sum_{p\in P}\sum_{t\in\mathcal{T}_p}s^{pd}_{p,d,t}\right) \tag{1}$$

S.T.:

Truck Movement

$$\sum_{a\in\delta^+(i)}x_{k,a}-\sum_{a\in\delta^-(i)}x_{k,a}=0,\quad \forall i\in N/\{(z_0,1),(z_0,T)\},\forall k\in K \tag{2}$$

$$\sum_{a\in\delta^+((z_0,1))}x_{k,a}=y_k,\quad \forall k\in K \tag{3}$$

$$\sum_{a\in\delta^-((z_0,T))}x_{k,a}=y_k,\quad \forall k\in K \tag{4}$$

Droid Movement

$$\sum_{a\in\delta^+(i)}(u_{d,a}+\sum_{k\in K}v_{d,k,a})-\sum_{a\in\delta^-(i)}(u_{d,a}+\sum_{k\in K}v_{d,k,a})=0,\quad \forall i\in N/\{(z_0,1),(z_0,T)\},\forall d\in D \tag{5}$$

$$\sum_{a\in\delta^+((z_0,1))}(u_{d,a}+\sum_{k\in K}v_{d,k,a})=w_d,\quad \forall d\in D \tag{6}$$

$$\sum_{a\in\delta^-((z_0,T))}(u_{d,a}+\sum_{k\in K}v_{d,k,a})=w_d,\quad \forall d\in D \tag{7}$$

$$\sum_{d\in D}v_{d,k,a}\le Q^t_{droid}\cdot x_{k,a},\quad \forall k\in K,\forall a\in A \tag{8}$$

Parcel Movement

$$\sum_{a\in\delta^+(i)}(\sum_{k\in K}f^r_{r,k,a}+\sum_{d\in D}g^r_{r,d,a})+h^r_{r,a^+_{stay}(i)}-\sum_{a\in\delta^-(i)}(\sum_{k\in K}f^r_{r,k,a}+\sum_{d\in D}g^r_{r,d,a})-h^r_{r,a^-_{stay}(i)}=0,$$
$$\forall i\in(Z\times\mathcal{T}_r)/\{(z_0,1),(\overline{z}_r,t):t\in\mathcal{T}_r\},\forall r\in\mathcal{R} \tag{9}$$

$$\sum_{a\in\delta^+((z_0,1))}(\sum_{k\in K}f^r_{r,k,a}+\sum_{d\in D}g^r_{r,d,a})+h^r_{r,a^+_{stay}((z_0,1))}=1,\quad \forall r\in\mathcal{R} \tag{10}$$

$$\sum_{a\in\delta^+((\overline{z}_r,t+1))}(\sum_{k\in K}f^r_{r,k,a}+\sum_{d\in D}g^r_{r,d,a})-\sum_{a\in\delta^-((\overline{z}_r,t+1))}(\sum_{k\in K}f^r_{r,k,a}+\sum_{d\in D}g^r_{r,d,a})=-(\sum_{k\in K}s^{rt}_{r,k,t}+\sum_{d\in D}s^{rd}_{r,d,t}),$$
$$\forall r\in\mathcal{R},\forall t\in\mathcal{T}_r \tag{11}$$

$$\sum_{r\in\mathcal{R}}f^r_{r,k,a}+\sum_{p\in P}f^p_{p,k,a}\le Q^t_{parcel}\cdot x_{k,a},\quad \forall k\in K,\forall a\in A \tag{12}$$

$$\sum_{r\in\mathcal{R}}g^r_{r,d,a}+\sum_{p\in P}g^p_{p,d,a}\le Q^d_{parcel}\cdot u_{d,a},\quad \forall d\in D,\forall a\in A \tag{13}$$

$$\sum_{a\in A/A(\overline{t}_r)}f^r_{r,k,a}=0,\quad \forall r\in\mathcal{R},\forall k\in K \tag{14a}$$

$$\sum_{a\in A/A(\overline{t}_r)}g^r_{r,d,a}=0,\quad \forall r\in\mathcal{R},\forall d\in D \tag{14b}$$

$$h^r_{r,a}=0,\quad \forall r\in\mathcal{R},\forall a\in A^{stay}/\{((z_0,t),(z_0,t+1)):t\in\mathcal{T}/\{T\}\} \tag{15}$$

Parcel Delivery

$$\sum_{t\in\mathcal{T}_r}\sum_{k\in K} s^{rt}_{r,k,t} + \sum_{t\in\mathcal{T}_r}\sum_{d\in D} s^{rd}_{r,d,t} = 1, \quad \forall r\in\mathcal{R} \tag{16}$$

$$s^{rt}_{r,k,t} \le \sum_{a\in a^+_{stay}((z_r,t))} f^r_{r,k,a}, \quad \forall k\in K, t\in\mathcal{T}_r, \forall r\in\mathcal{R} \tag{17}$$

$$\sum_{r\in\mathcal{R}:t\le\bar{t}_r} s^{rt}_{r,k,t} + \sum_{p\in P:t\ge\bar{t}_p} s^{pt}_{p,k,t} \le 1, \quad \forall k\in K, \forall t\in\mathcal{T} \tag{18}$$

$$s^{rd}_{r,d,t} \le \sum_{a\in a^+_{stay}((z_r,t))} g^r_{r,d,a}, \quad \forall d\in D, t\in\mathcal{T}_r, \forall r\in\mathcal{R} \tag{19}$$

$$\sum_{r\in\mathcal{R}:t\le\bar{t}_r} s^{rd}_{r,d,t} + \sum_{p\in P:t\ge\bar{t}_p} s^{pd}_{p,d,t} \le S^{max}, \quad \forall d\in D, \forall t\in\mathcal{T} \tag{20}$$

Parcel Transfer Limitations

$$\sum_{a\in\delta^-(i)} f^r_{r,k,a} + \sum_{a\in\delta^+(i)} f^r_{r,k',a} \le 1, \quad \forall r\in\mathcal{R}, \forall k,k'\ne k\in K, \forall i\in N \tag{21}$$

$$\sum_{a\in\delta^-(i)} f^p_{p,k,a} + \sum_{a\in\delta^+(i)} f^p_{p,k',a} \le 1, \quad \forall p\in P, \forall k,k'\ne k\in K, \forall i\in N \tag{22}$$

Droid Energy

$$\sum_{a\in\delta^-(i)} u_{d,a} + \sum_{k\in K}\sum_{a\in\delta^+(i)} v_{d,k,a} \le 1 + e^{reset}_{d,t}, \quad \forall d\in D, \forall i\in N \tag{23}$$

$$\sum_{a\in\delta^+(i)} u_{d,a} \le 1 - e^{reset}_{d,t}, \quad \forall d\in D, \forall i\in N \tag{24}$$

$$\begin{aligned} e^{counter}_{d,t} \le\ & M\cdot e^{reset}_{d,t} + e^{counter}_{d,t-1} - \sum_{a\in A:\, a=((\cdot,t-1),(\cdot,t))} e_a\, u_{d,a} \\ & - \sum_{r\in\mathcal{R}:\, t-1\le\bar{t}_r} s^{rd}_{r,d,t-1} - \sum_{p\in P:\, t-1\ge\bar{t}_p} s^{pd}_{p,d,t-1} \end{aligned} \qquad \forall d\in D,\ \forall t\in\mathcal{T}\setminus\{1\}, \tag{25a}$$

$$\begin{aligned} e^{counter}_{d,t} \ge\ & -M\cdot e^{reset}_{d,t} + e^{counter}_{d,t-1} - \sum_{a\in A:\, a=((\cdot,t-1),(\cdot,t))} e_a\, u_{d,a} \\ & - \sum_{r\in\mathcal{R}:\, t-1\le\bar{t}_r} s^{rd}_{r,d,t-1} - \sum_{p\in P:\, t-1\ge\bar{t}_p} s^{pd}_{p,d,t-1} \end{aligned} \qquad \forall d\in D,\ \forall t\in\mathcal{T}\setminus\{1\}. \tag{25b}$$

$$e^{counter}_{d,t} \le M\cdot(1-e^{reset}_{d,t}) + R, \forall d\in D, \forall t\in T/\{1\} \tag{26a}$$

$$e^{counter}_{d,t} \ge -M\cdot(1-e^{reset}_{d,t}) + R, \forall d\in D, \forall t\in T/\{1\} \tag{26b}$$

Pickup Parcel Movement

$$\sum_{a\in\delta^+(i)}\left(\sum_{k\in K} f^p_{p,k,a} + \sum_{d\in D} g^p_{p,d,a}\right) + h^p_{p,a^+_{stay}(i)} - \sum_{a\in\delta^-(i)}\left(\sum_{k\in K} f^p_{p,k,a} + \sum_{d\in D} g^p_{p,d,a}\right) + h^p_{p,a^-_{stay}(i)} = 0, \tag{27}$$
$$\forall i\in(Z\times\mathcal{T}_p)/\{(\bar{z}_{p,t},t) : t\in\mathcal{T}_p, (z_0,T)\},\ \forall p\in\mathcal{P}$$

$$\sum_{a\in\delta^+((\bar{z}_p,t))}\left(\sum_{k\in K} f^p_{p,k,a} + \sum_{d\in D} g^p_{p,d,a}\right) - \sum_{a\in\delta^-((\bar{z}_p,t))}\left(\sum_{k\in K} f^p_{p,k,a} + \sum_{d\in D} g^p_{p,d,a}\right) = \sum_{k\in K} s^{pt}_{p,k,t} + \sum_{d\in D} s^{pd}_{p,d,t}, \tag{28}$$
$$\forall p\in\mathcal{P}, \forall t\in\mathcal{T}_p$$

$$\sum_{a\in\delta^-((z_0,T))}\left(\sum_{k\in K} f^p_{p,k,a} + \sum_{d\in D} g^p_{p,d,a}\right) + h^p_{p,a^-_{stay}((z_0,T))} = 1, \forall p\in\mathcal{P} \tag{29}$$

$$\sum_{a \in A/A(\bar{t}_p)} f^p_{p,k,a} = 0, \quad \forall p \in \mathcal{P}, \forall k \in K \tag{30a}$$

$$\sum_{a \in A/A(\bar{t}_p)} g^p_{p,d,a} = 0, \quad \forall p \in \mathcal{P}, \forall d \in D \tag{30b}$$

$$h^p_{p,a} = 0, \quad \forall p \in \mathcal{P}, \forall a \in A^{stay}/\{((z_0,t),(z_0,t+1)) : t \in \mathcal{T}/\{T\}\} \tag{31}$$

Pickup Service

$$\sum_{t \in \mathcal{T}_p} \sum_{k \in K} s^{pt}_{p,k,t} + \sum_{t \in \mathcal{T}_p} \sum_{d \in D} s^{pd}_{p,d,t} = 1, \forall p \in P \tag{32}$$

$$s^{pt}_{p,k,t} \leq \sum_{a \in a^+_{stay}((\overline{z}_p,t))} f^p_{p,k,a}, \forall p \in P, \forall k \in K, \forall t \in \mathcal{T}_p \tag{33}$$

$$s^{pd}_{p,d,t} \leq \sum_{a \in a^+_{stay}((\overline{z}_p,t))} g^p_{p,d,a}, \forall p \in P, \forall d \in D, \forall t \in \mathcal{T}_p \tag{34}$$

Explanation

1. Objective function; minimize total cost
2. The number of flow arcs for truck k into node i should be equal to number of flow arcs out of node i.
3. If truck k is deployed, exactly 1 flow arc leaves the depot zone z_0 at $t = 1$ for truck k; together with constraint 2, this constraint enforces that a truck visits any node max. once.
7. If truck k is deployed, exactly 1 flow arc enters the depot zone z_0 at $t = T$ for truck k.
5. The number of flow arcs for droid d into node i should be equal to number of flow arcs out of node i.
6. If droid d is deployed, exactly 1 flow arc leaves the depot zone z_0 at $t = 1$ for droid d; together with constraint 5, this constraint enforces that a droid visits any node max. once.
7. If droid d is deployed, exactly 1 flow arc enters the depot zone z_0 at $t = T$ for droid d.
8. Enforces no more than Q^t_{droid} droids ride truck k on the same arc.
9. The number of flow arcs for delivery request r into node i should be equal to number of flow arcs out of node i.
10. Exactly 1 flow arc leaves the depot zone z_0 at $t = 1$ delivery request r; together with constraint 9, this constraint enforces that a parcel visits any node max. once.
11. If parcel r is serviced at time step t, exactly 1 flow arc enters node $(\overline{z}_r, t)$, and none leave that node.
12. Enforces no more than Q^t_{parcel} parcels ride truck k on the same arc.
13. Enforces no more than Q^d_{parcel} parcels ride droid d on the same arc, and only when droid d traverses the arc independently.
30. No parcel flow for request r after $\bar{t}_r$.
16. Any parcel should be delivered by max. one vehicle.
17. Parcel r can only be delivered by truck k at timestep t if the truck arrives at node $(\overline{z}_r, t)$.
18. Max. 1 delivery per truck per timestep.
19. Parcel r can only be delivered by droid d at timestep t if the droid arrives at node $(\overline{z}_r, t)$.
20. Max. S^{max} deliveries per droid per timestep.
23. Detects the moment the droid boards a truck after operating independently and reset to full energy.
24. Makes "reset" consistent with "boarding a truck", so the droid cannot depart independently while resetting.
25. Update $e^{counter}_{d,t}$: if $e^{rest}_{d,t} = 0$, subtract last period's energy use; if $e^{reset}_{d,t} = 1$, relax via M.
26. If $e^{reset}_{d,t} = 1$, force $e^{counter}_{d,t} = R$, otherwise relax via M.

27. Pickup flow conservation for pickup request p, the amount of pickup flow entering any intermediate node equals the amount leaving that node.

28. When pickup p is served at $(\underline{z}_{p,t}$, one unit of pickup flow is created at that node.

29. Exactly one unit of pickup flow for pickup p must end at the depot (z_0, T).

32. Each pickup request p must be served exactly once by either a truck or a droid, at an allowed pickup time.

33. Pickup p can only be served by truck k at time t if the pickup parcel is on the outgoing stay arc at $(\underline{z}_{p,t})$.

34 Pickup p can only be served by droid d at time t if the pickup parcel is on the outgoing stay arc at $(\underline{z}_{p,t})$.